Indian E-Commerce Sales & Customer Analytics

Dashboard Report | Data Analytics, Business Intelligence and Data Visualization

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Important: all data in this report is **synthetic** (generated with Python, Pandas, NumPy and Faker). It is not real company data and must not be read as real Indian e-commerce market statistics.

1. Introduction

This report presents the findings of an interactive analytics dashboard built on a simulated Indian e-commerce business covering January 2023 to December 2025. The dataset has 9 linked tables: 25,000 customers, 1,00,000 orders, 2,54,331 order items, 550 products, 1,00,000 payments, 89,681 shipments, 12,075 return records, 77,530 reviews and 151 marketing campaigns, across 15 states and 31 cities.

The dashboard answers seven groups of business questions: overall performance, sales trends, customer behaviour, product mix, geography, operations and returns, and payments and marketing. Every number below was recalculated from the source CSV files and cross-checked against the dashboard.

2. Executive KPI summary

Total order value (all statuses)	**₹257.26 Cr**	Delivered orders only	**₹208.43 Cr**
Total orders	**1,00,000**	Average order value	**₹25,726**
Registered customers	**25,000**	Customers who purchased*	**22,700**
Repeat / one-time customers	**18,695 / 4,005**	Never purchased	**2,300**
Delivered shipments late	**42.8%**	Return rate (orders)**	**10.9%**
Campaign-attributed revenue	**₹43.63 Cr**	Payments successful	**82.1%**

* A purchase = any order that is not Failed. ** Orders with at least one return record ÷ orders with status Delivered or Returned (9,510 ÷ 87,360). 1 Cr (crore) = 10 million rupees; 1 L (lakh) = 100,000.

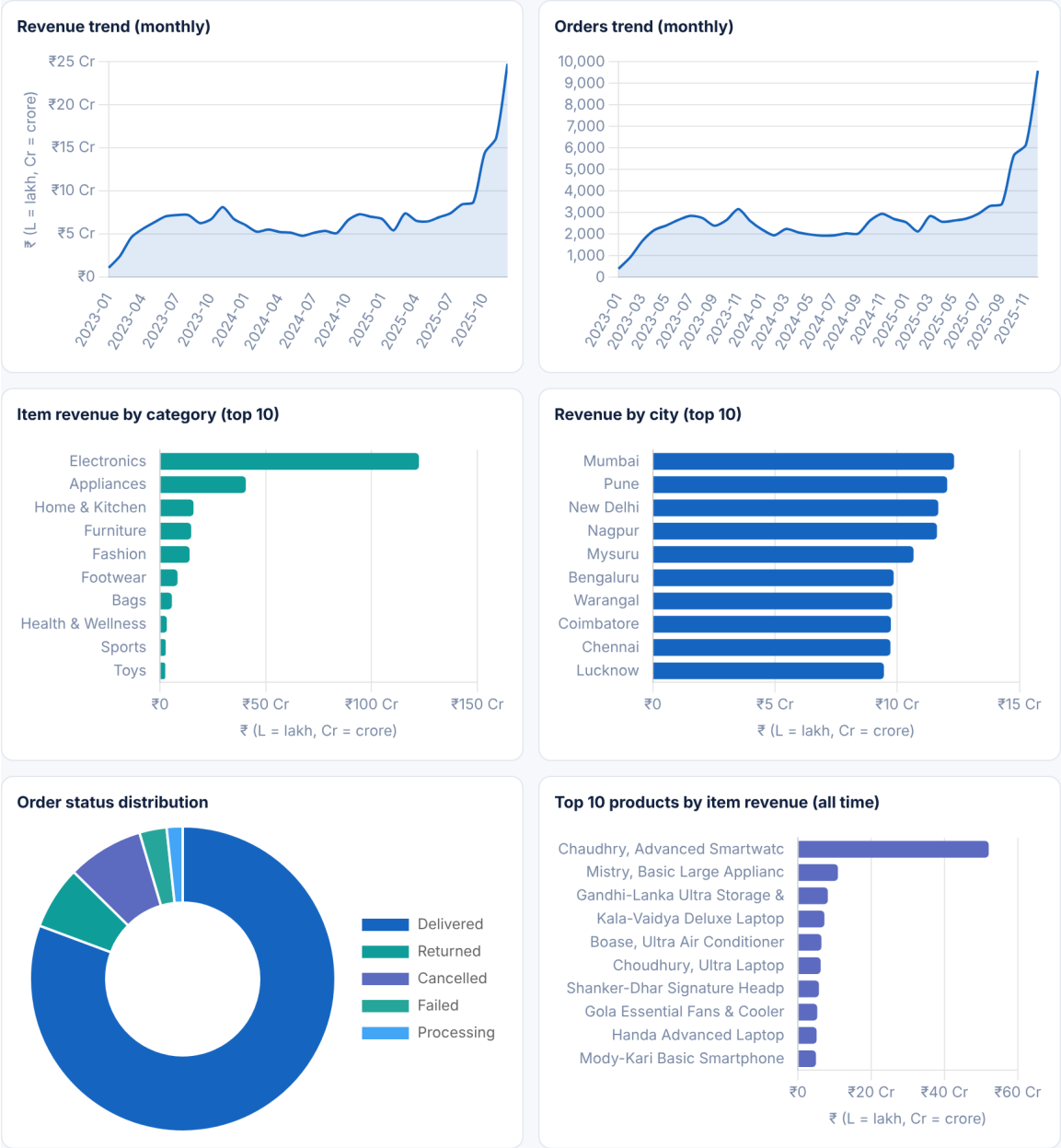
Why two revenue numbers?

Power BI "Total Revenue" sums **final_amount of every order** (₹257.26 Cr), including Failed (₹7.18 Cr), Cancelled (₹20.26 Cr) and Returned (₹16.99 Cr) orders. The earlier Python EDA left out Failed orders, which explains both the 2,880-order gap (97,120 + 2,880 Failed = 1,00,000) and the revenue gap (₹250.08 Cr + ₹7.18 Cr = ₹257.26 Cr). Details are on the last pages.

3. Executive Overview

Headline KPIs for the full period. (The interactive dashboard lets you filter by month, state, city, category and order status.)

₹257.26 Cr	1,00,000	₹25,726	₹208.43 Cr	25,000
Total revenue (selected)	Total orders	Average order value	Delivered revenue	Registered customers (all)



Key findings

- Total order value is ₹257.26 Cr from 1,00,000 orders, an average of ₹25,726 per order.
- Delivered orders make up 80.7% of orders and ₹208.43 Cr (81.0%) of value. Cancelled (8.1%), Returned (6.7%), Failed (2.9%) and Processing (1.7%) make up the rest.
- Electronics is about 50% of item revenue (₹122.36 Cr of ₹244.99 Cr), followed by Appliances (₹40.52 Cr).
- Mumbai (₹12.32 Cr) and Pune (₹12.04 Cr) are the top two cities.

Business takeaway

Revenue is concentrated in one category, so a supply or pricing problem in Electronics would hit the whole business. Always state which order statuses a revenue figure includes.

4. Sales Performance Analysis

Monthly, quarterly and year-over-year patterns, order volume and average order value.



Key findings

- Revenue by year: ₹69.41 Cr (2023), ₹68.48 Cr (2024), ₹119.38 Cr (2025). Orders: 26,750, 26,730 and 46,520.
- The 2025 jump (+74%) comes from more orders, not bigger orders: average order value stayed near ₹25.6K–25.9K in all three years.
- Q4 is the strongest quarter every year: 31.1% (2023), 30.5% (2024) and 46.4% (2025) of that year's revenue. December 2025 is the highest month at ₹24.78 Cr.
- January 2023 is the lowest month (₹1.07 Cr), the start of the simulated business.

Business takeaway

Plan stock, staff and marketing for a strong October to December peak. Because the data is synthetic, the 2025 growth should not be presented as real market growth.

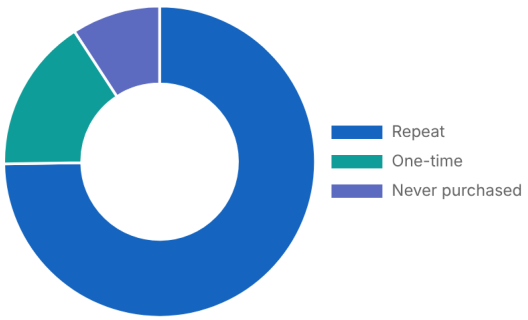
5. Customer Analytics

Customer types, purchase frequency, segments and new versus returning customers. A purchase means any non-Failed order; repeat means 2 or more such orders.

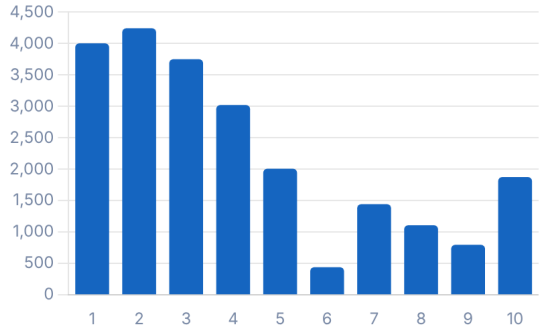
25,000	22,700	18,695	4,005	2,300
Registered customers	Purchasing customers	Repeat (2+ valid orders)	One-time	Never purchased

38.6%
Top 10% of buyers share of revenue

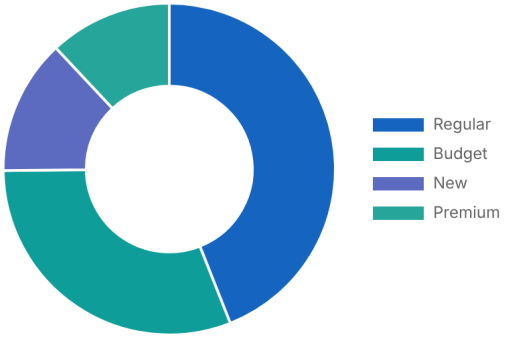
Customer type



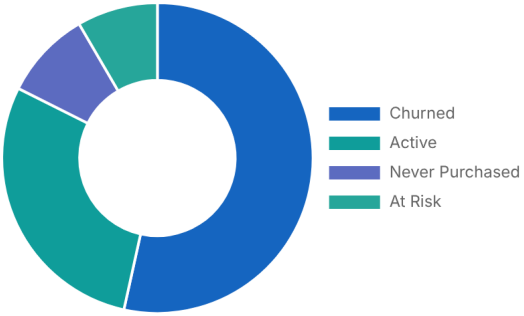
Purchase frequency (valid orders per buyer, 10 = 10+)



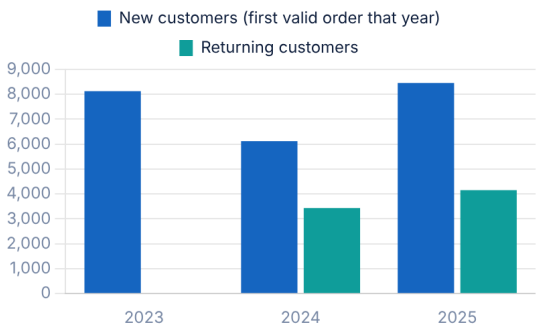
Customer segment (from dataset)



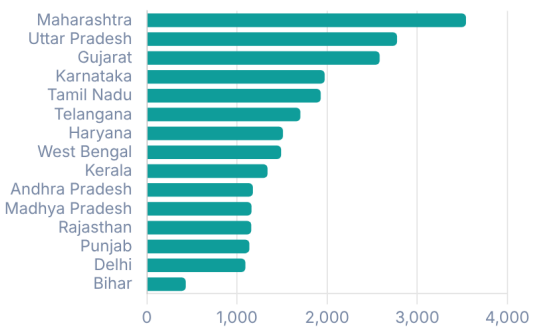
Customer status (from dataset)



New vs returning buyers by year (distinct customers)



Registered customers by state



Key findings

- 22,700 of 25,000 registered customers purchased: 18,695 repeat (82.4% of buyers), 4,005 one-time, and 2,300 never purchased (this includes 120 customers whose only orders Failed).
- The top 10% of buyers generate 38.6% of order value.
- 13,365 customers (53.5%) are labelled Churned, 7,234 Active and 2,101 At Risk in the dataset.
- New customers (first valid order in the year): 8,126 in 2023, 6,121 in 2024, 8,453 in 2025; returning customers: 3,438 in 2024 and 4,156 in 2025.

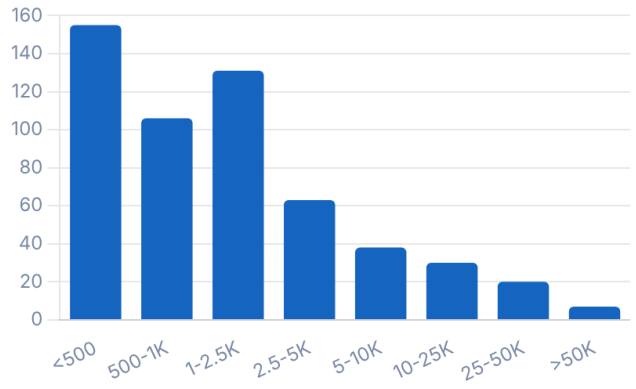
Business takeaway

High repeat behaviour is good, but the large Churned share suggests win-back campaigns and loyalty rewards for top spenders would pay off. No lifetime-value figure is claimed.

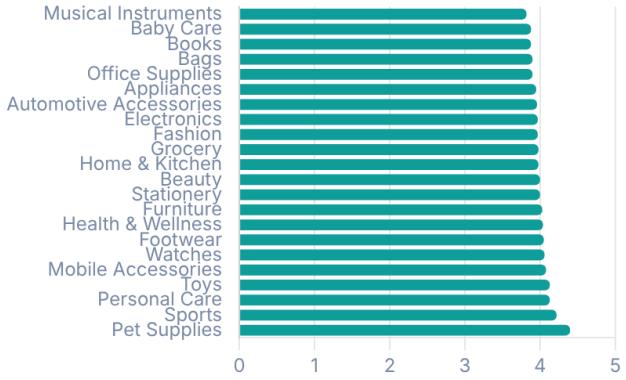
6. Product & Category Analysis

Price bands, ratings and the strongest and weakest products by item revenue (all statuses, from order items to avoid double counting).

Price band (number of products)



Average rating by category (product table)



Top 10 products

Product	Category	Revenue	Qty
Chaudhry, Advanced Smartwatche	Electronics	₹52.07 Cr	9,048
Mistry, Basic Large Appliance	Appliances	₹10.90 Cr	10,089
Gandhi-Lanka Ultra Storage & Container	Home & Kitchen	₹8.14 Cr	11,062
Kala-Vaidya Deluxe Laptop	Electronics	₹7.20 Cr	1,379
Boase, Ultra Air Conditioner	Appliances	₹6.39 Cr	1,543
Choudhury, Ultra Laptop	Electronics	₹6.18 Cr	2,222
Shanker-Dhar Signature Headphone	Electronics	₹5.71 Cr	8,774
Gola Essential Fans & Cooler	Appliances	₹5.25 Cr	1,020
Handa Advanced Laptop	Electronics	₹5.08 Cr	761
Mody-Kari Basic Smartphone	Electronics	₹4.91 Cr	1,270

Lowest 10 products

Product	Category	Revenue	Qty
Baral, Classic Bath & Body	Personal Care	₹3,934	36
Shanker-Dhar Signature Staple	Grocery	₹3,992	51
Roy-Sibal Everyday Dairy	Grocery	₹4,405	100
Sabharwal-Kothari Signature Snack	Grocery	₹5,819	76
Lanka, Essential Beverage	Grocery	₹7,248	115
Thaman-Sandhu Classic Phone Case	Mobile Accessories	₹8,069	60
Ramakrishnan Premium Pens & Pencil	Stationery	₹10,251	149
Lala, Deluxe Medical Device	Health & Wellness	₹13,258	63
Sarin Compact Oral Care	Personal Care	₹17,316	121
More, Essential Supplement	Health & Wellness	₹19,849	100

Key findings

- One product, "Chaudhry, Advanced Smartwatche", earns ₹52.07 Cr, about 21% of all item revenue.
- The top 10 products are mostly Electronics and Appliances. The bottom 10 are low-priced Grocery, Personal Care, Stationery and Health items with under ₹20K each.
- 155 of 550 products are priced under ₹500 and only 7 above ₹50,000.
- Item-level profit totals ₹19.73 Cr (8.1% of item revenue, all order statuses), so margins are thin.

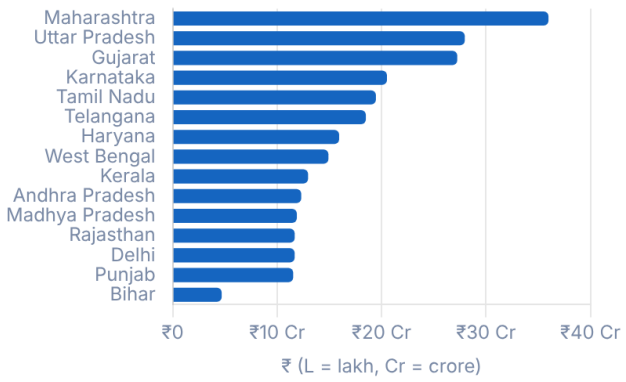
Business takeaway

Dependence on a few high-value products is a risk. Low sellers need a decision: promote, bundle or drop. A product this dominant is probably an artefact of the data generator.

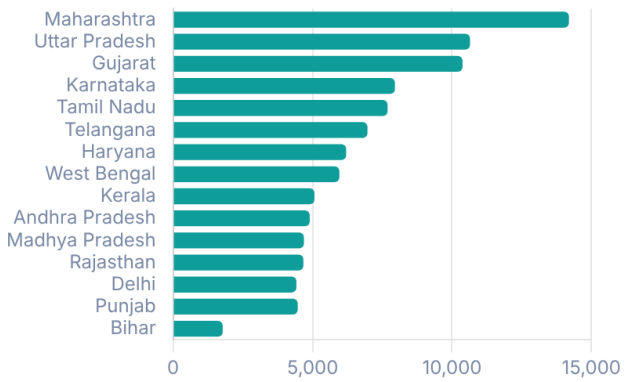
7. Geographic Analysis

Orders use the delivery state and city; customer counts use the registered state (these match on 93.6% of orders).

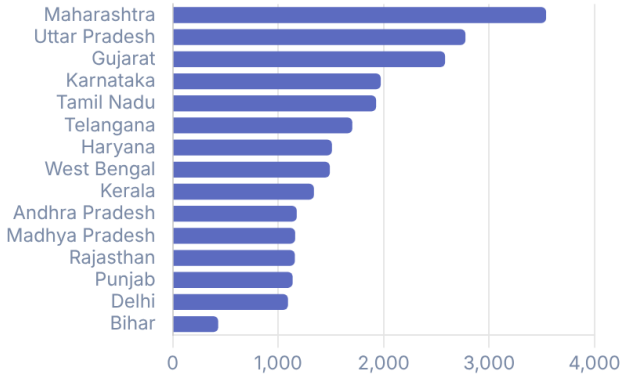
Revenue by state



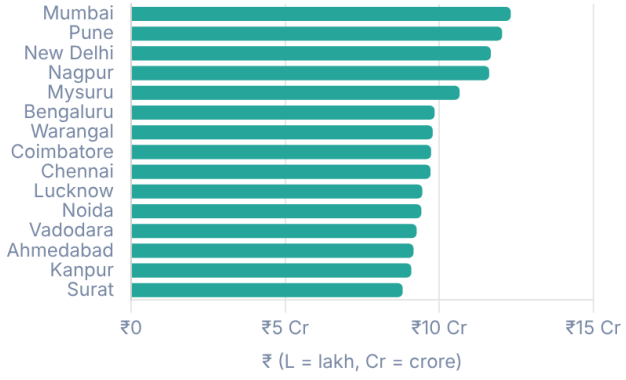
Orders by state



Registered customers by state (all time)



Revenue by city (top 15)



Key findings

- Maharashtra leads with ₹35.99 Cr (14.0% of order value), then Uttar Pradesh (₹27.97 Cr), Gujarat (₹27.25 Cr) and Karnataka (₹20.52 Cr).
- Top cities: Mumbai ₹12.32 Cr, Pune ₹12.04 Cr, New Delhi ₹11.68 Cr, Nagpur ₹11.62 Cr, Mysuru ₹10.66 Cr.
- Bihar is the smallest market: ₹4.69 Cr and only 431 registered customers.

Business takeaway

Sales are spread over 15 states, so no state is a single point of failure. Bihar shows where acquisition could grow; the top states deserve faster delivery and local offers.

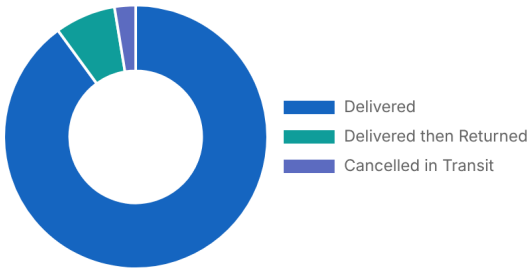
8. Operations, Delivery & Returns

Shipments, delay and return analysis.

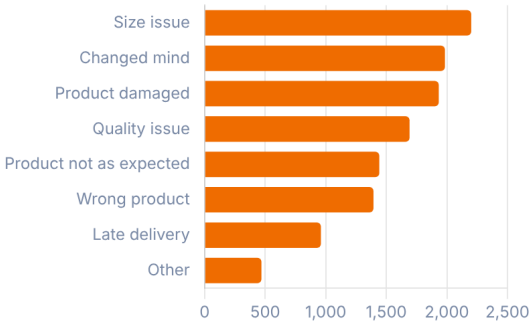
89,681	4.11	42.8%	12,075	10.9%
Shipments	Avg delivery days (delivered)	Delayed deliveries	Return records	Return rate (orders)

Definitions. Return rate = orders with at least one return record (9,510) ÷ delivered or returned orders (87,360). Delayed = actual delivery later than expected, among Delivered shipments. 910 delivery_days values do not equal actual minus dispatch date in the source, so I report the dataset values as given.

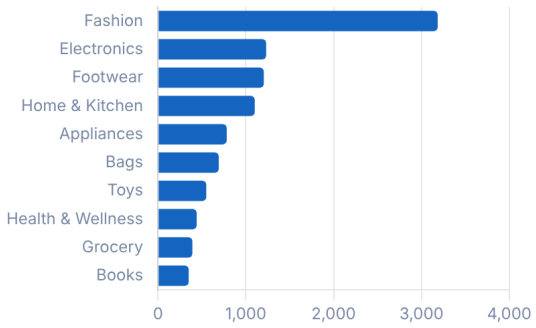
Shipment status



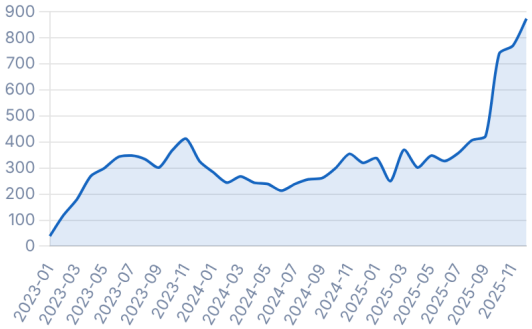
Return reasons



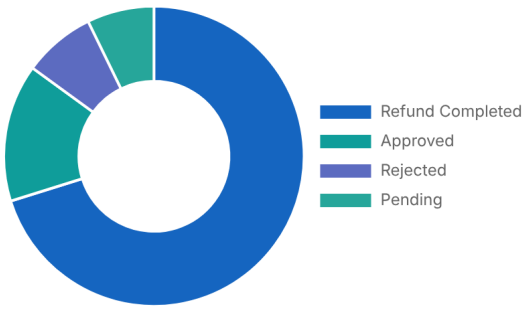
Returns by category (top 10)



Return records by month



Return request status



Key findings

- 89,681 shipments: 80,679 delivered, 6,681 delivered then returned, 2,321 cancelled in transit. Average delivery takes 4.11 days after dispatch.
- 42.8% of delivered shipments arrived after the expected date. By method: Same-Day 57.4% late, Standard 54.8%, Express 39.7%, Economy 38.3%, Next-Day 10.8%.
- 12,075 return records cover 9,510 orders (return rate 10.9%). Top reasons: Size issue (2,201), Changed mind (1,984), Product damaged (1,933).
- Fashion has the most return records (3,185). Footwear has the highest return records per item sold (8.5%), then Fashion (7.4%).
- Returns jump to 740, 770 and 874 per month in Oct to Dec 2025, versus about 300 in earlier months.

Business takeaway

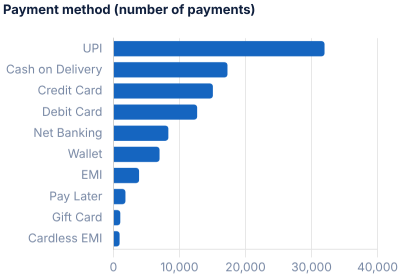
Review size guides for Fashion and Footwear, and check whether Same-Day and Standard promises are realistic. Treat late deliveries and the late-2025 return spike as the first things to investigate.

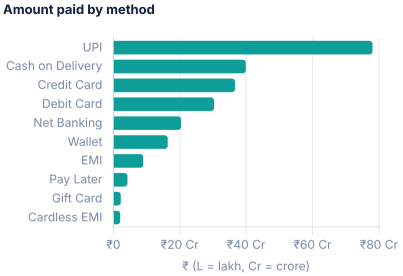
9. Payments & Marketing Analysis

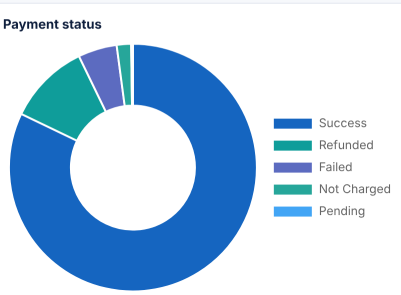
Payment methods and status, plus campaign-attributed performance. Attributed revenue is not incremental revenue, and the ratios below are not true ROI.

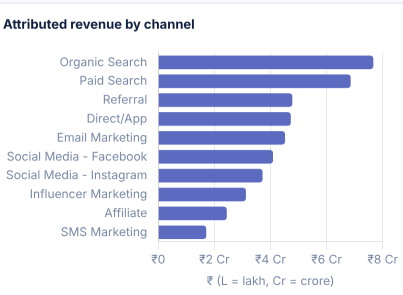
151	17,179 (17.2%)	₹43.63 Cr	₹18.71 Cr
Campaigns	Campaign-attributed orders	Attributed revenue	Campaign cost (dataset)

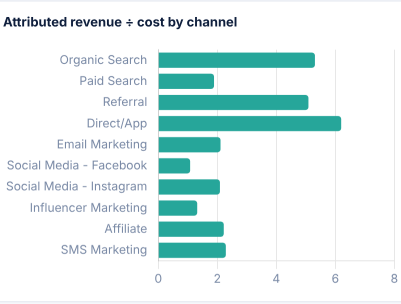
Careful. Attributed revenue is the total of orders tagged with a campaign. It is **not** incremental revenue (we do not know what those customers would have bought anyway), so this is not true marketing ROI. The revenue ÷ cost ratio below is simple, not profit.

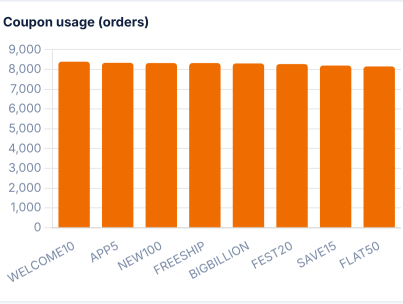


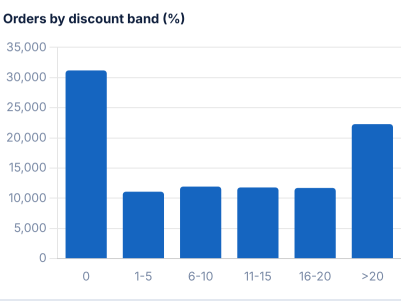












Key findings

- UPI is the most used method: 31,971 payments (32.0%) worth ₹78.06 Cr, ahead of Cash on Delivery (17,274) and Credit Card (15,063).
- 82,137 payments succeeded (82.1%), 10,690 were refunded, 5,068 failed, 1,876 not charged and 229 pending.
- 151 campaigns are tied to 17,179 orders (17.2% of orders) worth ₹43.63 Cr, against ₹18.71 Cr of campaign cost in the data (about 2.3x attributed revenue per rupee).
- Organic Search shows about 5.3x attributed revenue per rupee of cost; Facebook only about 1.1x. 66.3% of orders used a coupon, with average discount 12%.

Business takeaway

Shift budget toward channels with higher attributed revenue per rupee, then run a holdout test before calling any channel profitable.

10. Data checks and reconciliation

Structure checks passed: all primary keys unique, no orphan foreign keys, no duplicate rows, order subtotal equals the sum of item revenue, and final_amount equals subtotal + shipping + tax.

Resolved

- **2,880-order gap:** exactly the number of Failed orders. The customer table (and earlier EDA) excludes Failed orders: 97,120 + 2,880 = 1,00,000.
- **Revenue gap:** repeat ₹241.05 Cr + one-time ₹9.03 Cr = ₹250.08 Cr, which is ₹257.26 Cr minus Failed ₹7.18 Cr.

Not resolved (reported, not changed)

- 2,458 return records belong to orders with status Delivered; 6,681 orders have status Returned but 9,510 orders have return records.
- Refunded payments total ₹26.87 Cr versus ₹10.59 Cr in the returns table (4,009 cancelled orders also have refunded payments, but the gap is not fully reconciled).
- 910 shipments have delivery_days that differ from actual date minus dispatch date.
- Orders without shipment rows are Cancelled, Failed or Processing only (expected).

11. Executive summary

1. Revenue depends on the definition: ₹257.26 Cr (all orders) versus ₹208.43 Cr (delivered only).
2. Order volume, not order value, drove 2025: orders +74%, average order value flat.
3. Q4 is the peak quarter in every year.
4. Electronics is half of item revenue; one product is 21%.
5. 82.4% of buyers are repeat customers, yet 53.5% of customers are labelled Churned.
6. The top 10% of buyers deliver 38.6% of revenue.
7. Maharashtra, Uttar Pradesh and Gujarat lead; Bihar is the smallest market.
8. 42.8% of delivered shipments are late; returns spiked in late 2025, led by Fashion and Footwear.
9. UPI is the leading payment method; 82.1% of payments succeed.
10. Campaigns touch 17.2% of orders; this is attribution, not proven incremental revenue.

12. Conclusion

The dashboard turns nine linked tables into a single decision view. The main business messages are to protect the Electronics-heavy revenue base, prepare for the Q4 peak, win back churned customers, fix delivery promises and fashion returns, and test marketing channels properly. The main project lesson is that definitions matter: the same data gives ₹257.26 Cr or ₹208.43 Cr depending on which orders count, so every metric must state its filter. All results come from a synthetic dataset and demonstrate analytical method, not real market facts.